

A1: Floe-Property and Domain Sweep

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For which floes does the transmissibility inverse work?

Amir Dollie (DLLAMI004), EEE4022S

SHARC buoy project Code: `runA1FloePropertySweep.m, a1Tools.m` Results: `Results/A1/A1_20261003_091530`

Summary

Question. With the sensing held fixed (Level 2C at fractions of each floe’s true radius, reference s26, LSM6DSV16X, setting F), which floes can the inverse identify? 24 pre-registered scenarios: 8 one-parameter ring cases, 4 corners, 12 physical floes (basin, tank, SHARC model floe, MIZ). **Outcome.** 12 Identifiable, 6 Marginal, 6 Not identifiable (class robust to the EMM truncation, $\sigma_{\ln \beta}$ not converged). The domain has **two boundaries with different causes**. *Lower, physical:* below $R/\ell_f \approx 1$ the stiffness becomes unobservable ($\sigma_{\ln \beta}$ from 14% to 440%) while the radius stays measurable (0.4 to 1.7%); the cliff lies between $R/\ell_f = 0.96$ and 1.64 and is bracketed, not located. *Upper, estimator-limited:* at $kR \approx 3.7$ the precision is excellent ($\sigma_{\ln \beta} \approx 0.2\%$) but the Welch bias exceeds the class tolerance ($d_W \approx 0.6$). Between them, kR sets the precision. Stiffness itself matters little: β/β_0 from 0.25 to 4 moves $\sigma_{\ln \beta}$ only between 0.35 and 0.42%. The recorded model-floe hypothesis is confirmed ($\sigma_{\ln \beta}/\sigma_{\ln R} = 600$ to 900).

1 What the study does

Scaling. Each physical floe is converted with one water depth throughout: $D = Eh^3/(12(1 - \nu^2))$, $\beta = D/(\rho_w g H^4)$, $\gamma = \rho_i h/(\rho_w H)$, $R = R_{\text{phys}}/H$, with γ known and only $\theta = [\ln \beta, \ln R]$ estimated. Basin and tank floes use $H = 1.88$ m and the validated basin band (3.178 to 8.301 rad/s, $\alpha = 1.95$ to 13.15). The four MIZ floes use the effective depth of `screenFloeRegime` (122 m) and are evaluated *Froude-scaled* to the basin: the EMM sees only α, β, γ, R , so the 6 to 12.5 s band maps to basin frequencies, cut to the basin band (6.1 to 12.5 s, 69 bins). The equivalent full-scale experiment is $H_s = 3.25$ m, $T_p = 10.1$ s, 880 min, $\Delta t = 0.81$ s. Poisson’s ratio is passed to the EMM (0.40 for polypropylene).

Sensing held fixed. Sensors at the Level 2C fractions of each true R , converted once and held fixed while $\ln R$ is differentiated. The layout is *not* re-optimised per floe, so a change in performance comes from the floe, not from the layout.

Model validity before identifiability. Each scenario is checked before its class is read: (i) node grid 0.2 against 0.1 rad/s, feature error projected with the scenario’s own F , $d < 0.1$; (ii) truncation [50 10 10] against [70 15 15] as *metric* convergence: σ within 1%, d_W within 0.02, same class; (iii) all bins inside the validated α window. A failure is labelled *Model check failed*, never *Not identifiable*. The feature-level truncation bias is computed and reported for V1 but is not a gate. Rules (ii), the MIZ scaling and the ν handling are logged amendments made before the batch; rule (ii) replaced a feature-bias gate after the p_0 pre-flight (`diagnoseTruncationP0`): at p_0 the metrics moved by less than 0.05% while the feature bias was $d = 0.57$.

Checks. `testA1FloePropertySweep` 12/12 (the A1 path reproduces `evaluateScenario` exactly at p_0). One scenario (ring_beta_x0.25) first failed in the EMM’s damped-root solver; the root exists (continued smoothly in α), and a fallback that only runs when the original seed grid finds nothing recovered it (`testDampedRootFallback` 3/3, original roots unchanged).

2 Results

Parameter space (basin, $\gamma = \gamma_0$, $\nu = 0.3$; I = Identifiable, M = Marginal).

Scenario	β/β_0	R/R_0	kR_{pk}	R/ℓ_f	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	$\kappa(F)$	d_{trunc}	Class
ring β	0.25	1	1.84	6.54	0.385%	0.069%	0.317	111	0.69	I
ring β	0.5	1	1.84	5.50	0.354%	0.066%	0.306	99	0.67	I
p_0	1	1	1.84	4.63	0.355%	0.067%	0.309	99	0.57	I
ring β	2	1	1.84	3.89	0.375%	0.076%	0.289	118	0.49	I
ring β	4	1	1.84	3.27	0.421%	0.095%	0.248	146	0.38	I
ring R	1	0.5	0.92	2.31	0.953%	0.241%	0.112	83	0.30	I
ring R	1	0.75	1.38	3.47	0.622%	0.138%	0.187	197	0.53	I
ring R	1	1.5	2.75	6.94	0.221%	0.042%	0.455	86	0.50	I
ring R	1	2	3.67	9.25	0.187%	0.037%	0.617	66	1.57	M (bias)
corner	0.5	0.5	0.92	2.75	1.200%	0.300%	0.085	208	0.45	M (σ)
corner	0.5	2	3.67	11.0	0.197%	0.039%	0.593	49	2.97	M (bias)
corner	2	0.5	0.92	1.95	0.900%	0.201%	0.123	43	0.17	I
corner	2	2	3.67	7.78	0.171%	0.034%	0.587	79	0.58	M (bias)

Physical floes (N.I.[†]: Not identifiable at both truncations; the run’s label is *Model check failed* because $\sigma_{\ln \beta}$ changes by 1.5 to 31% between truncations, so its value is not quotable).

Scenario	Description	kR_{pk}	R/ℓ_f	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	d_W	d_{trunc}	Class
pp3mm_R072	PP 3 mm, $R = 0.72$ m (p_0 analogue)	1.84	5.06	0.348%	0.065%	0.305	0.63	I
pp5mm_D100	PP 5 mm, 1 m diameter	1.27	2.40	0.616%	0.158%	0.105	0.34	I
pp5mm_D40	PP 5 mm, 40 cm	0.51	0.96	(14%)	0.616%	0.073	0.43	N.I. [†]
pp10mm_D40	PP 10 mm, 40 cm	0.51	0.57	(121%)	0.690%	0.068	0.41	N.I. [†]
modelFloe	SHARC model floe, $E = 0.5/1.5/3$ GPa*	0.56	0.19–0.29	(240–440%)	0.40–0.57%	0.02	2.5–2.8	N.I. [†]
miz_R10_h1	$R = 10$ m, $h = 1$ m, 6 GPa	0.39	0.65	(111%)	1.71%	0.030	0.50	N.I. [†]
miz_R25_h1	$R = 25$ m, $h = 1$ m, 6 GPa	0.98	1.64	1.005%	0.184%	0.127	1.83	M (σ)
miz_R50_h1	$R = 50$ m, $h = 1$ m, 6 GPa	1.96	3.27	0.382%	0.086%	0.270	0.71	I
miz_R25_h05_E3	$R = 25$ m, $h = 0.5$ m, 3 GPa	0.98	3.27	1.198%	0.276%	0.106	0.83	M (σ)

*Provisional: 4 cm volume-equivalent plate of the 44 cm, 6 cm lens, E unknown. The basin p_0 row is in the parameter table.

Model checks. Node grid: all 24 pass, largest $d = 0.023$ (large- R cases). α : all inside [1.70, 13.85]. Truncation metric gate: 18 pass (largest change in σ 0.41%, in d_W 0.0022); the 6 failures are the N.I.[†] cases above, whose class is the same at both truncations. Feature-level truncation bias: $d_{trunc} \geq 0.5$ in **16 of 24** scenarios (largest 2.97 at the large- R corner; model floes 2.5 to 2.8).

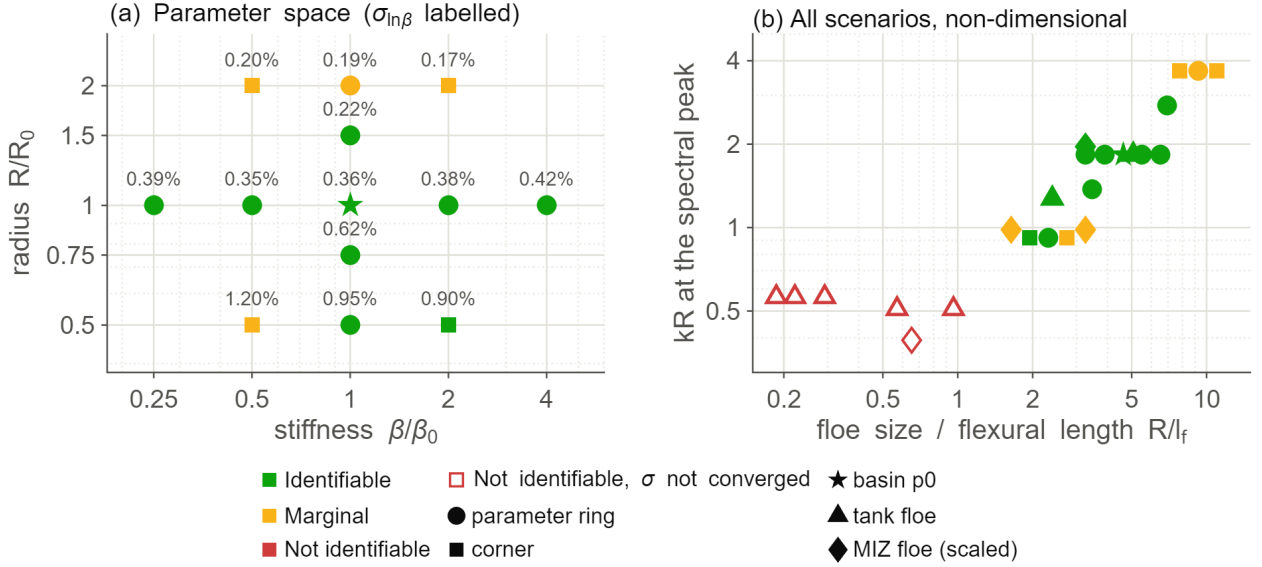


Figure 1. (a) Parameter-space scenarios, labelled with $\sigma_{\ln \beta}$. (b) All scenarios in the non-dimensional plane: floe size over flexural length against kR at the spectral peak. Colour: class; marker: group; open red: Not identifiable with σ not converged.

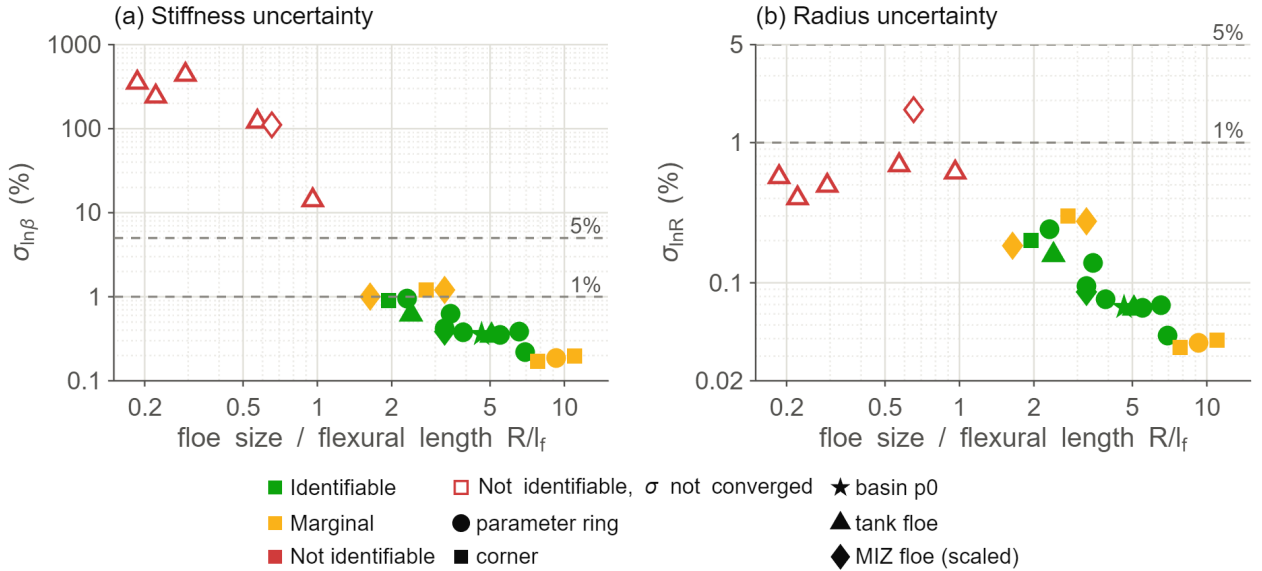


Figure 2. Stiffness (a) and radius (b) uncertainty against R/ℓ_f . Dashed: the 1% and 5% class limits. Below $R/\ell_f \approx 1$ the stiffness is lost; the radius is not.

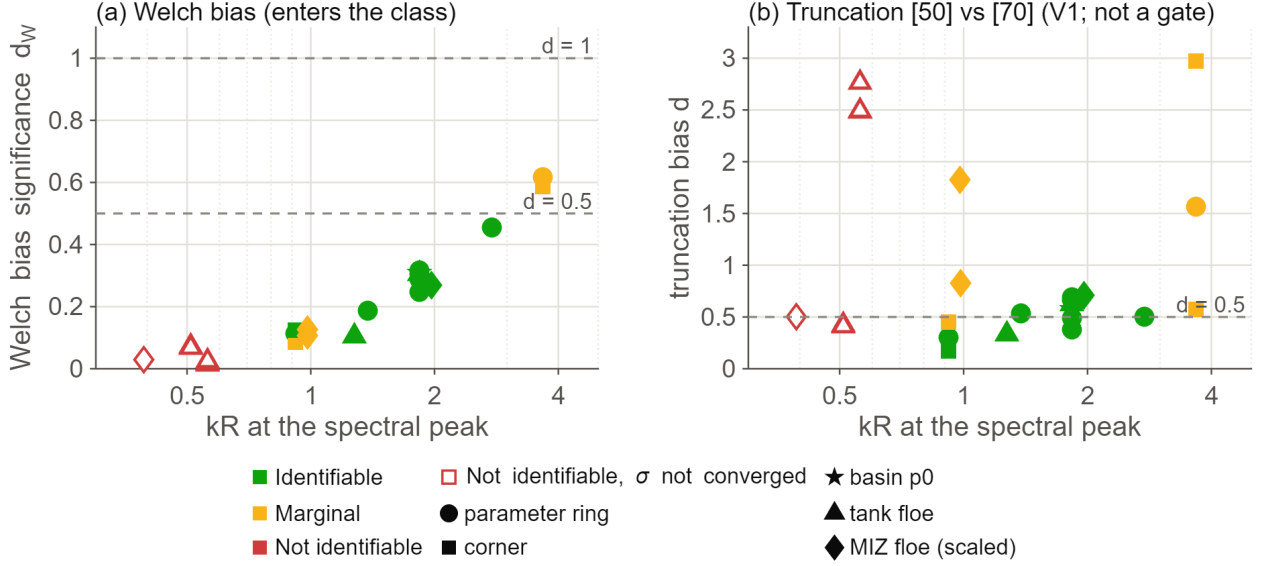


Figure 3. (a) Welch bias significance, which enters the class, against kR : it grows with kR and crosses 0.5 at the largest floes. (b) Feature-level bias of the [50 10 10] model against [70 15 15] (information for V1, not a gate).

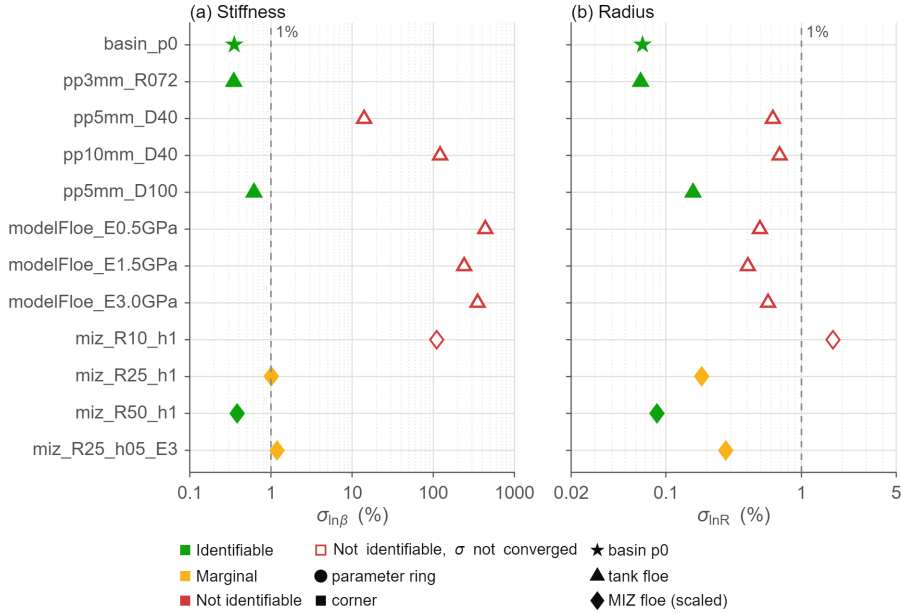


Figure 4. The physical floes: stiffness (a) and radius (b) uncertainty. Open markers: Not identifiable, value not converged.

3 Interpretation

- Lower boundary: the floe must be large compared with its flexural length.** Every scenario with $R/\ell_f \leq 0.96$ is Not identifiable in β , every one with $R/\ell_f \geq 1.64$ has $\sigma_{\ln \beta} \leq 1.2\%$. A floe much smaller than ℓ_f hardly bends in the band, so its motion carries little stiffness information; its size still shapes the transmissibility, so R stays measurable. No scenario lies between 0.96 and 1.64, so no threshold is claimed.
- Upper boundary: estimator-limited, not physical.** At $kR \approx 3.7$ ($R = 2R_0$ and its corners) the Fisher precision is the best in the study (0.17 to 0.20%), but $d_W = 0.59$ to 0.62 exceeds 0.5 and the class becomes Marginal. This is the mechanism of D2: σ falls while the window bias does not. D2 suggests that a longer L would recover these cases; that is a hypothesis for a logged diagnostic, not a result of A1.
- In between, kR sets the precision.** At $kR \approx 1$ $\sigma_{\ln \beta}$ is 0.9 to 1.2%, on the Identifiable/Marginal line; at kR 1.3 to 2 it is 0.35 to 0.6%. R/ℓ_f alone does not collapse the results: miz_R25.h05.E3 and miz_R50.h1 share $R/\ell_f = 3.27$ but have $kR = 0.98$ and 1.96 and $\sigma_{\ln \beta} = 1.20\%$ and 0.38%. R/ℓ_f decides whether flexural information exists; kR decides how much the sensors see.
- Stiffness itself is benign.** Over a 16-fold range of β at R_0 , $\sigma_{\ln \beta}$ stays between 0.35 and 0.42%, with a shallow minimum near $\beta_0/2$ to β_0 ; d_W falls slowly with β .
- Tank floes.** The 3 mm, 0.72 m polypropylene disc reproduces p_0 (0.348% against 0.355%), a check of the non-dimensional framework, and the 1 m disc is comfortably Identifiable. The 40 cm discs and the SHARC model floe are not: **the recorded hypothesis is confirmed**, the model floe is effectively rigid in the basin band ($\sigma_{\ln \beta}/\sigma_{\ln R} = 600$ to 900). A basin stiffness experiment needs a thin plate of about 1 m diameter.
- MIZ (scaled screening).** At $h = 1$ m, $E = 6$ GPa: 10 m Not identifiable, 25 m on the boundary (1.005%), 50 m Identifiable. A thinner, softer 25 m floe is Marginal for the same kR .

7. **The six robust Not identifiable cases** have weak stiffness information and an ill-conditioned weak direction (κ from 533 to 1.3×10^6). Their class does not depend on the truncation; their $\sigma_{\ln \beta}$ does, so only the class is reported.
8. **Converged metrics do not mean a converged model.** In 16 of 24 scenarios the [50 10 10] model differs from [70 15 15] by more than half a statistical uncertainty, although in every scenario that passes the metric gate σ and d_W move by less than 0.5%. The twin studies are unaffected (data and model share the EMM), but inverting *real* data with [50 10 10] would carry this bias, most for large kR . [70 15 15] is itself not converged at p_0 ($d = 0.25$ to [90 20 20]; vertical-mode truncation M dominates). This raises the priority of V1.

Limitations.

- **Fixed sensing.** Level 2C at fractions of R ; for the 40 cm floes two sensors are 4 cm apart, which 10 cm buoys cannot realise. The small-floe results describe the floe, not a buildable array.
- **One sea state and one direction** (JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s; R2, V2 and D3 extend this).
- **MIZ cases are Froude-scaled screening**, not a field prediction. A real LSM6DSV16X at full scale is about 8 times quieter in PSD than the scaled case assumes, so the MIZ results are conservative in noise.
- **Fixed basin band.** Changing R changes kR as well; A1 answers an engineering question (which floes work under this excitation band) rather than isolating floe properties.
- **Post-run reporting layer.** The “class robust” note on the six N.I.[†] cases was defined after the run; the run’s own label (Model check failed) is kept alongside.
- **Provenance.** 23 scenarios were computed under commit `d435bfc`, ring_beta_x0.25 under `a7a31df` (root fallback). The fallback is never reached for the other 23 and leaves the original roots unchanged.

Requirement for the experiment and the thesis

The inverse identifies stiffness for floes larger than about 1.6 flexural lengths and with kR of order 1 or more in the excitation band; below $R/\ell_f \approx 1$ only the size can be recovered. For a basin experiment, use a thin plate of about 1 m diameter (3 to 5 mm polypropylene); the SHARC model floe cannot yield a stiffness. For floes with large kR , the Welch resolution, not the information, limits the class. Before any real-data inversion, the EMM truncation must be raised and converged (V1). M1 boundary case (pre-registered rule): **miz R25 h1**.

Outputs (Results/A1/): A1_20261003.091530.log/.mat/_summary.txt; A1_scenarios.csv, A1_validity.csv, A1_equivalent.csv, A1_reporting.csv; checkpoints/; figures/A1_fig1 to A1_fig4 (.pdf/.png); diagnostics/truncationP0.*.